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Sperm whale depredation of sablefish longline gear in the northeast Pacific Ocean

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ABSTRACT

Interactions between marine mammals and fisheries include competition for prey (catch), marine mammal entanglement in fishing gear, and catch removal off fishing gear (depredation). We estimated the magnitude of sperm whale depredation on a major North Pacific longline fishery (sablefish) using data collected during annual longline surveys. Sperm whale depredation occurs while the longline gear is off-bottom during retrieval. Sperm whales were observed on 16% of longline survey sampling days, mostly (95% of sightings) over the continental slope. Sightings were most common in the central and eastern Gulf of Alaska (98% of sightings), occasional in the western Gulf of Alaska and Aleutian Islands, and absent in the Bering Sea. Longline survey catches were commonly preyed upon when sperm whales were present (65% of sightings), as evidenced by damaged fish. Neither sperm whale presence ($P = 0.71$) nor depredation rate ($P = 0.78$) increased significantly from 1998 to 2004. Longline survey catch rates were about 2% less at locations where depredation was observed, but the effect was not significant ($P = 0.34$). Estimated sperm whale depredation was <1% of the annual sablefish longline fishery catch off Alaska during 1998 to 2004.

Key words: sperm whale, *Physeter macrocephalus*, depredation, sablefish, longline fishery.

Interactions between marine mammals and commercial fisheries occur in many parts of the world (Yano and Dahlheim 1995, Ashford *et al.* 1996, Hucke-Gaete *et al.* 2004). Interactions include competition for prey (catch), marine mammal entanglement in fishing gear, and removal of catch off fishing gear (depredation). Sperm whale (*Physeter macrocephalus*) depredation of longline catches has been reported for Alaska and the southern oceans (Hill *et al.* 1999, Nolan and Liddle 2000, Hucke-Gaete *et al.* 2004, Purves *et al.* 2004). In Alaska, sperm whales prey upon sablefish (*Anoplopoma fimbria*), which are caught using longline gear. The Alaska sablefish fishery primarily is hook-and-line, the gear is fished on-bottom, and 30 to 40 million circle-type hooks are deployed each year (Sigler *et al.* 2003). Fishermen report that when sperm whales are present, sablefish catches sometimes decrease. In addition, some fish are damaged in characteristic ways, including missing body parts, crushed tissue, and blunt tooth marks. Damage occurs during gear retrieval, when sperm whales have been detected on the vessel depth finder located off-bottom and adjacent to the longline and on underwater video with their mouth around the line knocking caught sablefish from hooks (J. Straley, personal observation). Our goal was to estimate the magnitude of sperm whale depredation in the Alaska sablefish longline fishery and to determine whether the depredation rate has increased in recent years. Sperm whales also appear to consume fishery offal, as they often remain astern of vessels, where discarded fish parts (head and internal organs) are found.

Sperm whales are distributed across the North Pacific Ocean. Females and their dependent young usually remain in tropical and temperate waters year round, whereas males are thought to move north in summer to feed in the Gulf of Alaska, Bering Sea, and waters around the Aleutian Islands, returning south of 40°N during the winter (Angliss and Lodge 2004). Sperm whales feed primarily on cephalopods (Okutani and Nemoto 1964, Berzin 1971, Gaskin 1982, Rice 1989, Santos *et al.* 1999), but there are regions where fish is the predominant component of their diet (Berzin 1971, Clarke and Macleod 1976, Kawakami 1980, Gosho *et al.* 1984, Rice 1989). Kawakami (1980) reviewed sperm whale diets worldwide and found that fish were an important part of their diet in some areas of the North Pacific and North Atlantic Oceans and New Zealand waters. According to Gaskin (1982), the diet of pelagic populations of sperm whales includes more cephalopods, whereas coastal populations predominantly forage on fish. Stomach samples from specimens examined at whaling stations from whales caught off Alaska revealed that cephalopods were important food in the western Aleutian Islands and Bering Sea but that fish became progressively more important toward the eastern Aleutian Islands and into the Gulf of Alaska (Okutani and Nemoto 1964). Fish ingested in northern latitudes commonly included sablefish (Anoplomatidae, *Anoplopoma*) as well as skates (Rajidae, *Raja*), sharks (Chondrichthyes), sea devils (Ceratiidae), cod and hake (Gadidae, *Gadus* and *Merluccius*), grenadiers (Macrouridae, *Coryphaenoides* and *Albatrossia*), ribbonfishes (Trachipteridae, *Trachipterus*), ragfishes (Icosteidae, *Icosteus*), rockfishes (Scorpaenidae, *Sebastes*), and lingcod (Hexagrammidae, *Ophiodon*) (Rice 1989).

Data on sperm whale depredation were collected during annual sablefish longline surveys conducted by the National Marine Fisheries Service (NMFS) in coastal waters off Alaska from 1998 to 2004. The sablefish longline surveys have been conducted annually since 1978 to measure sablefish relative abundance (Sigler and Fujioka 1988, Sigler and Zenger 1994). Our objectives were to estimate the magnitude of sperm whale depredation on sablefish longline survey catches, to determine whether the depredation rate has increased in recent years, and to estimate the amount of sablefish longline fishery catch affected by sperm whale depredation.

Methods

Data Collection

Data on sperm whale depredation were collected during annual NMFS sablefish longline surveys from 1998 to 2004. Sablefish longline surveys are conducted in the Gulf of Alaska each year and in the southeast Bering Sea and the Aleutian Islands region in alternate years. The survey area generally covers the commercial fishing area, which is subdivided into six management areas for sablefish: Bering Sea, Aleutian Islands, Western Gulf of Alaska, Central Gulf of Alaska, West Yakutat, and Southeast (Fig. 1). Standard locations (stations) occur 30–60 km apart along the continental slope and in deep water cross-shelf gullies, sampling the seafloor at water depths from 150 to 900 m. In addition, one station in the Southeast area is located on the continental shelf. Typically, one station consisting of 7,200 hooks is sampled each day. The longline gear is set in the morning. Soak time ranges from 3 to 10 h and varies depending on the length of time needed to retrieve the gear. Hooks are spaced 2 m apart and are baited with squid. The longline gear is weighted with 3 kg weights every 100 m. Sperm whale depredation is assumed to have occurred when sperm whales are present (typically 100 m or less from the vessel) and sablefish damaged in characteristic ways (missing body parts, crushed tissue, blunt tooth marks, shredded bodies, lips remaining on hooks) are retrieved. It seems reasonable to classify this damage as due to sperm whales because sperm whales were observed under the vessel on the vessel fish finder when this characteristic damage was observed on a few occasions, the blunt teeth of sperm whales likely would cause the crushed tissue and

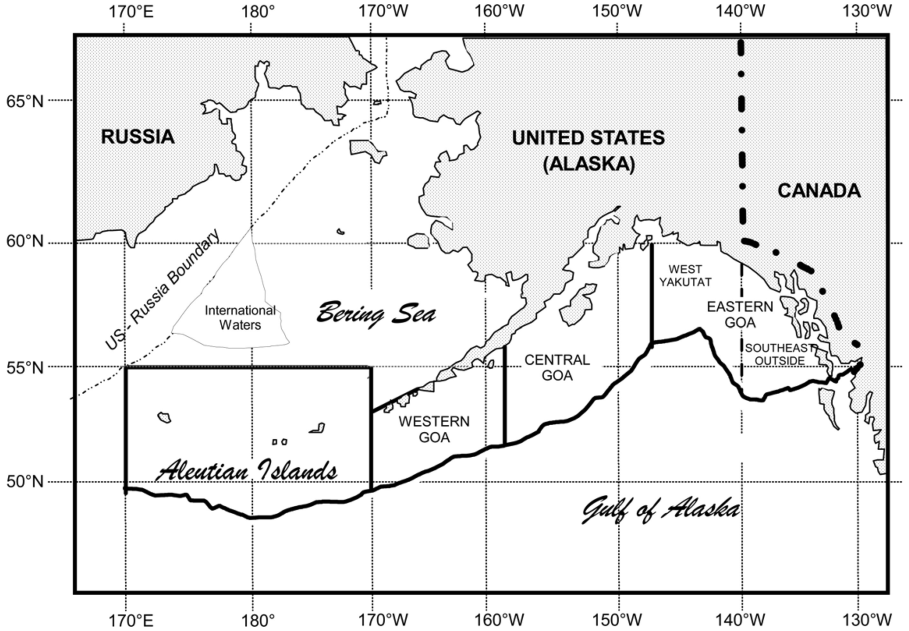

Figure 1. Map of sablefish management areas of the U.S. Exclusive Economic Zone off Alaska. The Gulf of Alaska is abbreviated as "GOA" in the map.

blunt tooth marks observed, and the sharp teeth of other potential predators such as sharks would result in cut tissue with well-defined borders. Information recorded includes the number of sperm whales present and the number of damaged sablefish retrieved.

Analysis

Data were stratified by sablefish management area. Frequency of occurrence of sperm whales was calculated as the number of sampling days when sperm whales were observed divided by the total number of sampling days. Frequency of occurrence of sperm whale depredation was calculated as the number of sampling days when sperm whale depredation was observed divided by the number of sampling days when sperm whales were observed. The percentage of damaged sablefish retrieved per sampling day was computed as the number of damaged sablefish divided by the total number of sablefish retrieved.

The effect of sperm whale depredation on sablefish catches was analyzed by comparing sablefish catch rates for stations where depredation was observed to stations where no depredation was observed. The average catch rate for a station was computed as the average sablefish catch rate (kg per hook) by depth strata weighted by the size of each depth strata (Sigler and Fujioka 1988). Only the area east of 159°W where sperm whale depredation was observed was analyzed (Central Gulf of Alaska, West Yakutat, and Southeast sablefish management areas), which encompasses four International North Pacific Fisheries Commission (INPFC) statistical areas: Chirikof, Kodiak, Yakutat, and Southeast (Sigler and Zenger 1994).

We tested for a statistically significant depredation effect by fitting a general linear mixed model (GLMM) and then constructing a contrast hypothesis test (Jennrich 1995). A repeated measures model was used to account for within-subject correlation (Littel *et al.* 1998). Each of 35 longline survey stations (subjects) had seven annual observations of sablefish catch rate from 1998 to 2004. The within-station correlation structure was modeled as heterogeneous autoregressive lag 1 and was selected on the basis of the Akaike Information Criterion (AIC) and residual plots. Autoregressive lag 1 means that the catch rates in successive years are correlated. Heterogeneous means that this autoregressive parameter differs between INPFC statistical areas. Each of the four INPFC statistical areas has a variance parameter as well as an autoregressive correlation parameter, thus introducing eight variance–covariance parameters that are needed to correctly estimate standard errors and P values for other model effects. Heteroscedasticity of the catch rate data was reduced through reiterated weighting (Kutner *et al.* 2004). Weights were set proportional to the reciprocal of squared residuals. The parameter estimates were stabilized and residual plots were satisfactory after three iterations of fitting, weighting, and refitting. We found that depredation was not a significant effect within the best-fitting model.

Depredation, although not a significant effect, is included in the model in order to construct the contrast hypothesis test and estimate the depredation effect. This hypothesis test controls for the within-station correlation by year, as well as the area effect. A nonparametric approach, bootstrap analysis, was completed in addition to the parametric approach of the contrast hypothesis test. The bootstrap method was to randomly resample the residuals from the best-fitting model (Manly 1997), adding the sampled residuals to the expected values of the catch rates for the best-fitting model and then estimating the depredation effect to create a distribution of 3,000

Table 1. Number of days sperm whales (SW) were present and total number (#) of sampling days each year for all years combined (1998–2004) by management area and habitat type.

Management area	Gully SW present	Gully #Sample days	Shelf SW present	Shelf #Sample days	Slope SW present	Slope #Sample days	Total SW present	Total #Sample days	Total Percent
Bering Sea	0	0	0	0	0	57	0	57	0%
Aleutian Islands	0	0	0	0	1	64	1	64	2%
Western Gulf of Alaska	0	0	0	0	1	70	1	70	1%
Central Gulf of Alaska	1	63	0	0	18	112	19	175	11%
West Yakutat	1	14	0	0	31	56	32	70	46%
Southeast	1	21	1	7	30	77	32	105	30%
Total	3	98	1	7	81	436	85	541	16%
Percent	3%		14%		19%				

depredation contrasts. All analyses were done with Statistical Analysis Software (SAS ver. 9.1).[1]

The annual amount of sablefish taken by sperm whale depredation of sablefish longline fishery catches was estimated. Total longline fishery catch before depredation for management area i ($C_{T,i}$) was estimated based on the percentage of sampling days that depredation was observed for area i (o_i), the percentage that catch was reduced when depredation was observed (d), and the observed average longline fishery catch (1998–2004) for area i ($C_{o,i}$): $C_{T,i} = C_{o,i}/(1 - o_i \cdot d)$. The value of d was estimated in the contrast hypothesis test. The amount of sablefish catch taken by sperm whale depredation (D_i) was obtained by subtraction: $D_i = C_{T,i} - C_{o,i}$.

Results

Sperm whales were observed on 16% of longline survey sampling days (85 of 541, Table 1). Most sightings (95%, 81 of 85) occurred over the continental slope. Sperm whales were observed on 19% of slope sampling days but only 3% of gully sampling days. Sperm whales were observed only once during seven survey years at the single continental shelf station. Most sightings of sperm whales occurred in the Central Gulf of Alaska, West Yakutat, and Southeast areas (98%, 83 of 85; Table 1). Sperm whales were more common in the West Yakutat (46% of sampling days) and Southeast areas (30%) than the Central Gulf of Alaska area (11%). Sperm whales only occasionally were observed in the Aleutian Islands and Western Gulf of Alaska areas and never were observed in the Bering Sea area. The remaining results focus on the slope habitat where most sperm whale sightings occurred and the sablefish longline fishery primarily occurs.

Depredation of sablefish longline survey catch was common when sperm whales were present. Damaged sablefish were retrieved at 65% of the locations where sperm

[1] Reference to trade names does not imply endorsement by the National Marine Fisheries Service, NOAA.

Table 2. Number of sampling days with sperm whales present (SW) and number of sampling days with sperm whale depredation (SWD) by management area and habitat type on the continental slope by year.[a]

		1998		1999		2000		2001		2002		2003		2004		Total	
Management area	ST	SW	SWD	SW	SWD	SW	SWD	SW	SWD	SW	SWD	SW	SWD	SW	SWD	SW	SWD
Bering Sea	19	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Aleutian Islands	16	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0
Western Gulf of Alaska	10	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0
Central Gulf of Alaska	16	1	0	4	3	0	0	4	3	3	3	2	2	4	3	18	14
West Yakutat	8	6	4	6	6	4	4	2	2	5	4	1	1	7	4	31	25
Southeast	11	4	0	4	4	5	2	2	2	5	2	2	2	8	2	30	14
Total		11	4	14	13	9	6	8	7	14	9	5	5	20	9	81	53
Percentage of sampling days that sperm whales were present		18%		22%		15%		13%		23%		8%		33%			
Percentage of days depredation was observed when sperm whales were present			36%		93%		67%		88%		64%		100%		45%		

[a]Number of sampling days (ST) per year by area; all areas are sampled annually, except two which are sampled alternate years, the Bering Sea (odd-numbered years) and Aleutian Islands (even-numbered years).

whales were present on the continental slope (53 of 81, Table 2). Sperm whale depredation was observed in the Central Gulf of Alaska, West Yakutat, and Southeast areas, but not in the Aleutian Islands or the Western Gulf of Alaska areas where sperm whales occasionally were observed. Few damaged sablefish were retrieved where sperm whale depredation was observed. An average of 7 and a maximum of 25 damaged sablefish were retrieved per sampling day where depredation was observed. The percentage of sablefish catch retrieved damaged also was small; the average percentage was 0.8% ± 0.14% (1 SE, $n = 55$) and the maximum percentage was 5.9% where depredation was observed.

The occurrence of sperm whales varied markedly from year to year. For the continental slope, sperm whales were observed at 8% (minimum) of longline survey stations in 2003 and 33% (maximum) of stations in 2004 (Table 2). The percentage of stations where sperm whales were observed did not change significantly from 1998 to 2004 (linear regression, df = 6, $P = 0.71$). The occurrence of sperm whales was unrelated to average sablefish catch rate (Fig. 2). The occurrence of sperm whale depredation also varied markedly among years, ranging from 36% (minimum) in 1998 to 100% (maximum) in 2003 (Table 2). The percentage of stations where sperm whale depredation was observed did not change significantly from 1998 to 2004 (linear regression, df = 6, $P = 0.78$).

Sablefish catch rates were less at longline survey stations where depredation was observed by −0.985 kg per 100 hooks based on the contrast hypothesis test, but the difference was not statistically significant (one-tailed t test, df = 21, SE = 2.31,

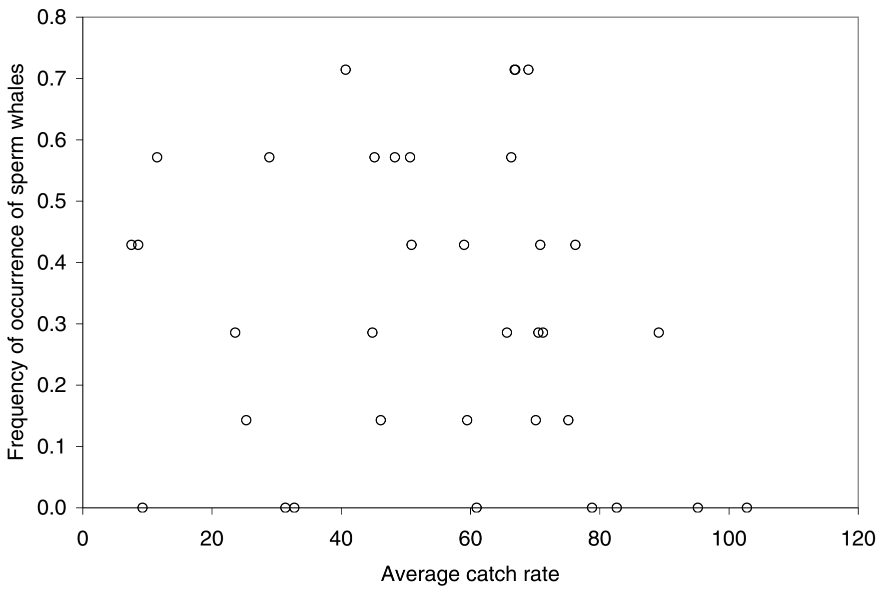

Figure 2. The frequency of occurrence of sperm whales versus the average catch rate (kg per 100 hooks) of sablefish by station, compiled from 1998 to 2004. Only the area east of 159°W where sperm whale depredation was observed was plotted (Central Gulf of Alaska, West Yakutat, and Southeast sablefish management areas), which encompasses four statistical areas: Chirikof, Kodiak, Yakutat, and Southeast.

Table 3. By management area, the percentage of days that depredation was observed when sperm whales were present (Table 2), the percent reduction in catch when sperm whale depredation was observed, the 5-yr average sablefish catch (*t*), the estimated total catch before depredation (*t*), and the estimated amount of sablefish catch lost to sperm whale depredation (*t*).

Management area	Percent of days	Catch reduced	Average catch	Estimated total catch	Estimated depredation
Bering Sea	0%	0.0%	454	454	0
Aleutian Islands	0%	0.0%	916	916	0
Western Gulf of Alaska	0%	0.0%	1,567	1,567	0
Central Gulf of Alaska	13%	0.2%	4,911	4,922	11
West Yakutat	45%	0.8%	1,738	1,752	14
Southeast	18%	0.3%	3,412	3,423	11
Total			12,997	13,034	36

$P = 0.34$). The 95% confidence interval for the depredation effect is (−5.80, 3.83). The bootstrap estimates are comparable, implying that the estimates are insensitive to the statistical method. The 95% confidence interval is (−5.12, 4.16) and the estimated catch rate reduction is (−0.384, median). The catch rate reduction based on the contrast hypothesis test implies sperm whales removed about 1.8% (d) of the sablefish at stations where depredation was observed. A preliminary analysis of a subset of the data (1998–2001) (Sigler *et al*. 2001) estimated a larger value ($d = 23\%$) comparing medians (depredation observed *vs.* not observed) of catch rate residuals by area. However, the preliminary analysis was less sophisticated than the current study analysis; for example, the statistically significant within-station correlation structure found in this study was not modeled. Thus we consider this study's estimate of depredation rate (d) more reliable and applied this value to estimate sperm whale depredation of sablefish longline fishery catches.

The amount of sablefish taken by sperm whale depredation of sablefish longline fishery catches appears small. The estimated amount was based on the percentage of longline survey sampling days that depredation was observed by area, the percent catch was reduced when depredation was observed ($d = 1.8\%$), and the observed average longline fishery catch (1998–2004) by management area. The estimated amount of sablefish taken during the commercial fishery is 36 t/yr, equivalent to 0.3% of the average annual longline fishery catch of 13,034 t (Table 3).

Discussion

Sperm whales prey upon sablefish caught on longline gear, but the overall effect appears minimal. Sperm whale depredation accounted for <1% of the annual sablefish longline fishery catch off Alaska. Neither sperm whale presence nor depredation rate increased significantly during annual sablefish longline surveys from 1998 to 2004.

A small depredation effect may be difficult to detect. The power of the contrast hypothesis test to detect the observed 1.8% reduction is 7% (power curve for t test, df = 21, SE = 2.31). A depredation effect would have to be greater than 8% to be found significant at $P = 0.05$ with this method. One previous study, from data collected by a fisheries observer program in Alaskan waters, also found no significant effect (Hill *et al.* 1999). The earlier study compared longline fishery catches between sets with sperm whales present and absent but did not record whether or not depredation was observed. Their analysis included all sperm whale sightings, whether depredation was observed or not, which would tend to mask any depredation effect. Another previous study, from data collected only in the Southeast area, found a small, significant effect comparing longline fishery catches between sets with sperm whales present and absent (3% reduction, t test, 95% CI 0.4%–5.5%, $P = 0.02$, Straley *et al.* 2005).

Some potential biases exist in our data. One potential bias exists because longline survey data were used to estimate sperm whale depredation for the longline fishery. Fishermen may employ strategies to reduce depredation (avoid areas of known sperm whale activity, depart areas when sperm whales are encountered), whereas longline survey areas are sampled regardless of sperm whale activity. On the other hand, fishermen often fish an area for a few days and sperm whales may accumulate around the vessel, whereas the survey vessel samples a different station each day. Another potential bias exists because sablefish may be removed without evidence of depredation (damaged fish or lips remaining on the hooks). We assumed some damaged sablefish would be observed when depredation occurred because sablefish are forcefully removed from hooks on the fishing vessel and presumably by sperm whales (most sablefish are hooked through a gill cover, which is a bony tissue). These potential biases appear small. Depredation estimates for sablefish longline data are similar (1%–3%) regardless whether from survey (our study) or fishery data (Hill *et al.* 1999, Straley *et al.* 2005) or whether classified for analysis based on evidence of depredation (our analysis) or sperm whale sighting alone (Hill *et al.* 1999, Straley *et al.* 2005) (Fig. 3).

At some stations, sperm whales were present, but no depredation was observed. These sperm whales presumably fed upon fishery offal (discarded heads and internal organs) instead. These sperm whales also may be feeding normally and by chance are at the same location as the longline survey vessel. Sperm whale depredation may be higher when fishery offal is unavailable. Anecdotally, fishermen report that sperm whale depredation is more common when sablefish are not headed and gutted at sea.

A study of sperm whale depredation off southern Chile on Patagonian toothfish (*Dissostichus eleginoides*) reported that 3% of fish (±2% CI 95%; $n = 180$) (TDLC in Hucke-Gaete *et al.* 2004) were retrieved damaged, somewhat higher than the 0.8% retrieved damaged that we report here for sablefish. Hucke-Gaete *et al.* (2004) also found that sperm whales were more common in areas where toothfish catch rates were higher, which they hypothesized occurred if the richest fishing grounds are also traditional feeding grounds for sperm whales. We found that the occurrence of sperm whales was unrelated to sablefish catch rate at the station level, but across a broader region, sperm whales were more common in the Central Gulf of Alaska, West Yakutat, and Southeast areas, where sablefish catch rates are higher (Sigler *et al*. 2003) and sperm whales have traditionally fed (Okutani and Nemoto 1964).

The effect of sperm whale depredation on longline fishery catches appears minor. Including our study, five studies of sperm whale depredation have measured catch rate reductions of only 1%–3% (Fig. 3). The overlapping confidence intervals of the catch rate reductions suggest that these estimates are not significantly different from one other. Only two of the five studies found the effect to be statistically significant (Hucke-Gaete *et al.* 2004, Straley *et al.* 2005). The five studies cover two major longline fishery species (sablefish and Patagonian toothfish).

Although the impact on fishery yield is low, sperm whale depredation remains a concern. Sperm whales may become entangled in the longline gear, suffering injury or

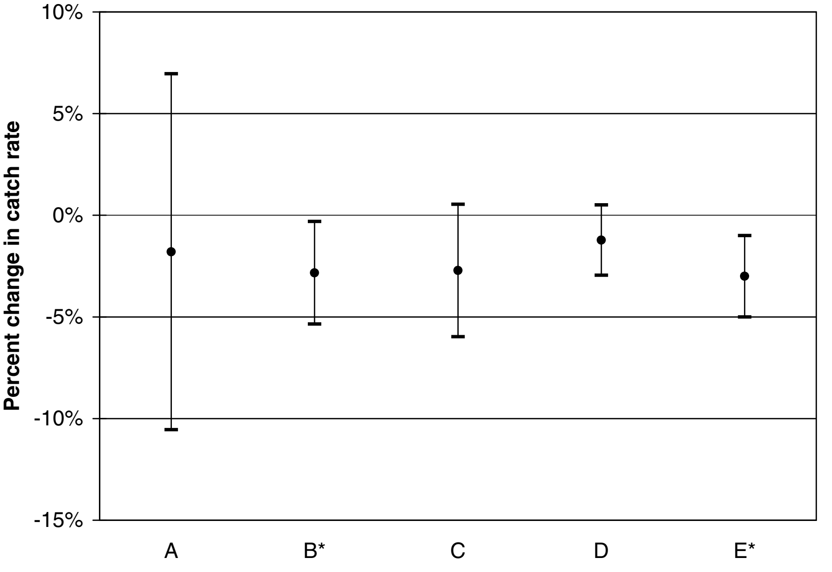

Figure 3. This plot compares sperm whale depredation estimates (percent change in catch rate) from five different studies (A: sablefish, Alaska, this study; B: sablefish, Alaska, Straley *et al.* 2005; C: sablefish, Alaska, Hill *et al.* 1999; D: Patagonian toothfish, South Georgia, Purves *et al.* 2004; and E: Patagonian toothfish, Chile, Hucke-Gaete *et al.* 2004). Hill *et al.* reported depredation in metric tons, which we translated into a percent reduction for this plot. The confidence interval for Purves *et al.* (2004) was estimated from information provided in the article; the hypothesis test was inverted to obtain an equivalent *t*-distribution confidence interval. Hucke-Gaete *et al.* (2004) did not report catch rate reduction but rather only the percentage of fish that were retrieved damaged, which is displayed in the figure. The asterisks indicate significant effects.

death as a result. The financial loss to fishermen may be significant because of the high market value of the fish (both sablefish and Patagonian toothfish are valuable) and the cumulative effect of operational interactions incurred when the vessel reacts to the presence of sperm whales (Hucke-Gaete *et al.* 2004). Depredation may affect some fishermen more than others so that some individual fishermen are greatly impacted even though the average loss is small. Other factors such as the availability of fishery offal may affect depredation rate and should be studied. Focal studies of sperm whale depredation (Straley *et al.* 2005) may lead to development of methods to reduce sperm whale interactions with longline vessels. Observations of sperm whale depredation of sablefish longline catches should continue in order to document any changes in the depredation rate.

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